

CO 2 ASSESSMENT TOOL – 2

DSA0134 – Object Oriented Programming with C++

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Scenario 1: City Traffic Management System:

```
#include <iostream>

#include <string>

using namespace std;

class Violation {
    string vehicleNum, owner, type;
    double fine;
    static int totalViolations;
public:
    void inputDetails() {
        cout << "Enter Vehicle Number, Owner, and Type (Speeding/Parking/Signal): ";
        cin >> vehicleNum >> owner >> type;
        if (type == "Speeding") fine = 1000.0;
        else if (type == "Parking") fine = 500.0;
        else fine = 1500.0;
        totalViolations++;
    }
    void display() {
        cout << vehicleNum << " | " << owner << " | " << type << " | $" << fine << endl;
    }
    static void showTotal() {
        cout << "Total Violations: " << totalViolations << endl;
    }
};

int Violation::totalViolations = 0;
```

```

int main() {
    Violation v[2];
    for(int i = 0; i < 2; i++) v[i].inputDetails();
    cout << "\n--- Violation Reports ---\n";
    for(int i = 0; i < 2; i++) v[i].display();
    Violation::showTotal();
    return 0;
}

```

OUTPUT:

Output

Clear

```

Enter Vehicle Number, Owner, and Type (Speeding/Parking/Signal): TN01 John
Speeding
Enter Vehicle Number, Owner, and Type (Speeding/Parking/Signal): KA05 Jane
Parking

--- Violation Reports ---
TN01 | John | Speeding | $1000
KA05 | Jane | Parking | $500
Total Violations: 2

=== Code Execution Successful ===

```

2. Fitness Centre Membership System

```

#include <iostream>
#include <string>
using namespace std;
class Member {
    int id;
    string name, plan;
    double fee;
    static int totalMembers;
public:

```

```

void assignPlan() {
    cout << "Enter ID, Name, and Plan (Basic/Premium/VIP): ";
    cin >> id >> name >> plan;
    if (plan == "Basic") fee = 30.0;
    else if (plan == "Premium") fee = 60.0;
    else fee = 100.0;
    totalMembers++;
}

void display() {
    cout << "ID: " << id << " | Name: " << name << " | Plan: " << plan << " | Fee: $" << fee
<< endl;
}

static void showTotal() {
    cout << "Total Active Members: " << totalMembers << endl;
}

};

int Member::totalMembers = 0;

int main() {
    Member m[2];
    for(int i = 0; i < 2; i++) m[i].assignPlan();
    cout << "\n--- Membership Details ---\n";
    for(int i = 0; i < 2; i++) m[i].display();
    Member::showTotal();
    return 0;
}

```

OUTPUT:

Output

```
Enter ID, Name, and Plan (Basic/Premium/VIP): 101 Alice Premium
Enter ID, Name, and Plan (Basic/Premium/VIP): 102 Bob Basic

--- Membership Details ---
ID: 101 | Name: Alice | Plan: Premium | Fee: $60
ID: 102 | Name: Bob | Plan: Basic | Fee: $30
Total Active Members: 2

=== Code Execution Successful ===
```

3. Logistics Company Warehouse:

```
#include <iostream>
#include <string>
using namespace std;
class Warehouse {
    int id, qty;
    string name, location;
    static int totalItems;
public:
    void addInventory() {
        cout << "Enter Item ID, Name, Quantity, and Location: ";
        cin >> id >> name >> qty >> location;
        if(qty > 0) totalItems += qty;
        else qty = 0;
    }
    void updateInventory(int change) {
        if (qty + change >= 0) {
            qty += change;
            totalItems += change;
        }
    }
};
```

```

    }

    void display() {
        cout << id << " | " << name << " | Qty: " << qty << " | Loc: " << location << endl;
    }

    static void showTotal() {
        cout << "Total Warehouse Items: " << totalItems << endl;
    }
};

int Warehouse::totalItems = 0;

int main() {
    Warehouse w[2];
    for(int i = 0; i < 2; i++) w[i].addInventory();
    cout << "\nRemoving 5 units from the first item...\n";
    w[0].updateInventory(-5);
    cout << "\n--- Inventory Status ---\n";
    for(int i = 0; i < 2; i++) w[i].display();
    Warehouse::showTotal();
    return 0;
}

```

OUTPUT:

Output

```

Enter Item ID, Name, Quantity, and Location: 501 Laptops 50 A1
Enter Item ID, Name, Quantity, and Location: 502 Monitors 30 A2

Removing 5 units from the first item...

--- Inventory Status ---
501 | Laptops | Qty: 45 | Loc: A1
502 | Monitors | Qty: 30 | Loc: A2
Total Warehouse Items: 75

=== Code Execution Successful ===

```

4. Event Registration System

```
#include <iostream>

#include <string>

using namespace std;

class Event {
    int regID;

    string eventName, participant;

    static int totalParticipants;

public:
    int getID() { return regID; }

    void registerParticipant(int id, string name, string eName) {
        regID = id;
        participant = name;
        eventName = eName;
        totalParticipants++;
    }

    void display() {
        cout << "ID: " << regID << " | " << participant << " | " << eventName << endl;
    }

    static void showTotal() {
        cout << "Total Registered: " << totalParticipants << endl;
    }
};

int Event::totalParticipants = 0;

int main() {
    Event events[2];

    int count = 0, id;

    string name, eName;

    while(count < 2) {
        cout << "Enter Reg ID, Participant Name, and Event: ";
```

```

    cin >> id >> name >> eName;

    bool dup = false;

    for(int i = 0; i < count; i++) {
        if(events[i].getID() == id) dup = true;
    }

    if(dup) cout << "Duplicate ID! Try again.\n";

    else events[count++].registerParticipant(id, name, eName);
}

cout << "\n--- Registration Details ---\n";

for(int i = 0; i < 2; i++) events[i].display();

Event::showTotal();

return 0;
}

```

OUTPUT:

Output

Clear

```

Enter Reg ID, Participant Name, and Event: 1001 Mark Hackathon
Enter Reg ID, Participant Name, and Event: 1001 Sarah Seminar
Duplicate ID! Try again.
Enter Reg ID, Participant Name, and Event: 1002 Sarah Seminar

--- Registration Details ---
ID: 1001 | Mark | Hackathon
ID: 1002 | Sarah | Seminar
Total Registered: 2

=== Code Execution Successful ===

```

5. E-Commerce Shopping Cart

```

#include <iostream>

#include <string>

using namespace std;

class Product {
    string name;

```

```

double price;

int qty;

static int accounts;

public:
    Product() {
        if(accounts == 0) accounts = 1;
    }
    void getData() {
        cout << "Enter Product Name, Price, and Quantity: ";
        cin >> name >> price >> qty;
    }
    double calcTotal() { return price * qty; }
    void display() {
        cout << name << " | $" << price << " | Qty: " << qty << " | Total: $" << calcTotal() <<
endl;
    }
    static void showAccounts() {
        cout << "Active Accounts: " << accounts << endl;
    }
};

int Product::accounts = 0;

int main() {
    Product cart[2];
    double grandTotal = 0;
    for(int i = 0; i < 2; i++) {
        cart[i].getData();
        grandTotal += cart[i].calcTotal();
    }
    cout << "\n--- Shopping Cart Bill ---\n";
    for(int i = 0; i < 2; i++) cart[i].display();
    cout << "-----\nGrand Total: $" << grandTotal << endl;
}

```

```
Product::showAccounts();  
return 0;  
}
```

OUTPUT:

Output

Clear

```
Enter Product Name, Price, and Quantity: Keyboard 45.50 2  
Enter Product Name, Price, and Quantity: Mouse 25.00 1  
  
--- Shopping Cart Bill ---  
Keyboard | $45.5 | Qty: 2 | Total: $91  
Mouse | $25 | Qty: 1 | Total: $25  
-----  
Grand Total: $116  
Active Accounts: 1  
  
=== Code Execution Successful ===
```